

MA341

APPLIED ODE'S

Handout 1

Definition 1. We say that n^{th} order differential equation

$$F\left(x, y, y', y'', \dots, y^{(n)}\right) = 0$$

is **linear** if it possible to rewrite it as following

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + a_{n-2}(x)y^{(n-2)} + \dots + a_1(x)y' + a_0(x)y = b(x).$$

Definition 2. A **general solution** to ODE is a solution that contains at least one arbitrary constant.

Definition 3. A **private solution** to ODE is obtained from the general solution by determining the values to all arbitrary constant.

Definition 4. Consider first order linear equation

$$y' + p(x)y = g(x).$$

If $p(x) = 0$ then y is an antiderivative of $g(x)$, i.e. $y = \int g(x)dx$. Otherwise one solves it with the help of **integration factor**

$$\mu = e^{\int_0^x p(t)dt},$$

to get

$$u' = (\mu y)' = \mu(y' + p(x)y) = \mu g(x) \equiv G(x).$$

Thus

$$u = \int G(x)dx,$$

which leads to

$$y = \frac{1}{\mu(x)} \left(\int \mu(x)g(x)dx + C \right).$$

If $p(x) = \alpha$ is a constant one get $\mu = e^{\alpha x}$, consequently

$$y = e^{-\alpha x} \left(\int e^{\alpha x}g(x)dx + C \right).$$